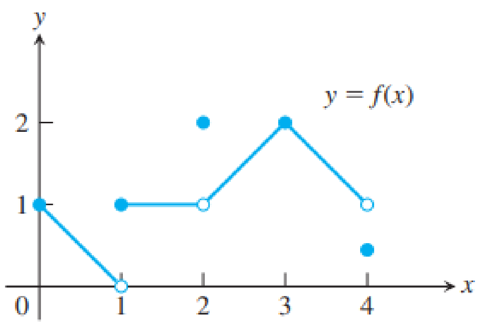
2.5 Continuity

When we plot function values, we often connect the plotted points with an unbroken curve to show what the function's values are likely to have been at the points we did not measure. In doing so, we are assuming that we are working with a *continuous function*, so its outputs vary regularly and consistently with the inputs, and do not jump abruptly from one value to another without taking on the values in between. Intuitively, any function $y = f(x)$ whose graph can be sketched over its domain in one unbroken motion is an example of a continuous function. Such functions play an important role in the study of calculus and its applications.

① Continuity at a Point

To understand continuity, it helps to consider a function like that in the following Figure, whose limits we investigated in Example 3 in the last section.

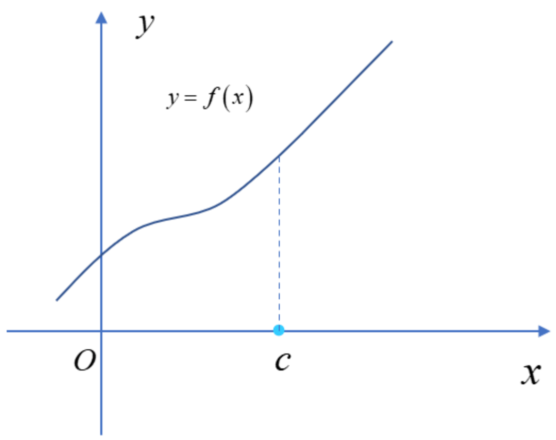


First we observe that the domain of the function is the closed interval [0, 4], so we will be considering the numbers x within that interval. From the figure, we notice right away that here are breaks in the graph at the numbers $x = 1$, $x = 2$, and $x = 4$. These are the discontinuous points.

The following definitions capture the continuity ideas we observed in the above.

- A function $f(x)$ is continuous at $x = c \in (a, b)$ $\Leftrightarrow$ $\lim_{x \to c} f(x) = f(c)$.

For example,



■ If a function is not continuous at an interior point c of its domain, we say that f is discontinuous at c, and that c is a point of discontinuity of f.

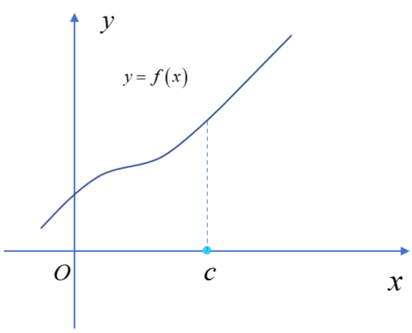
■ To determine if a function is continuous at c, we need to check if it satisfies the following three conditions:

(a) $f(x)$ is defined at $x = c$.

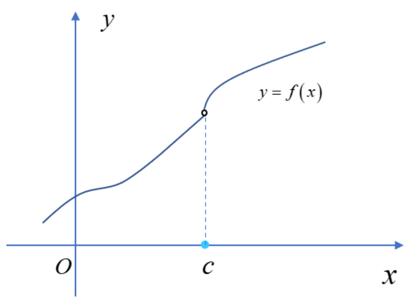
(b) $\lim_{x \to c} f(x)$ exists.

(c) $\lim_{x \to c} f(x) = f(c)$.

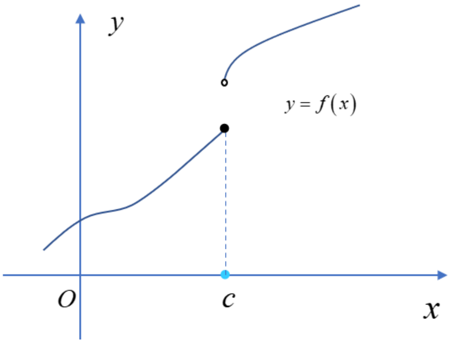
Note 1 If one of the above three conditions is false, then we say that $f(x)$ is discontinuous at $x = c$.



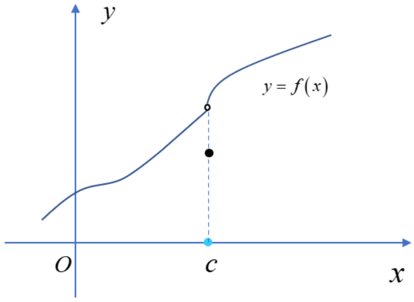
(a) $f(x)$ is continuous at c



(b) $f(x)$ is discontinuous at c



(c) $f(x)$ is discontinuous at c

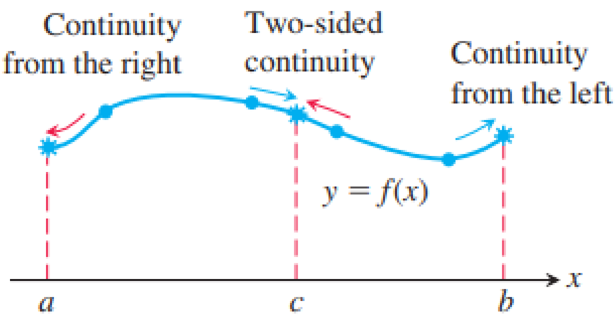


(d) $f(x)$ is discontinuous at c

Note 2 (a) A function $f(x)$ is right-continuous at $x = c \iff \lim_{x \to c^+} f(x) = f(c)$.

(b) A function $f(x)$ is left-continuous at $x = c \iff \lim_{x \to c^-} f(x) = f(c)$.

We say that a function is continuous over a closed interval $[a, b]$ if it is right-continuous at a, left-continuous at b, and continuous at all interior points of the interval.



Example 1 The function $f(x) = \sqrt{4 - x^2}$ is continuous over its domain $[-2, 2]$.

Example 2 Let

$$f(x) = \begin{cases} \dfrac{x^2 - x - 6}{x - 3}, & x \neq 3; \\ 5, & x = 3. \end{cases}$$

Show that $f(x)$ is continuous at $x = 3$.

Example 3 Show that $f(x)=\begin{cases} \dfrac{|x|}{x}, & x\neq 0; \\ 0, & x=0. \end{cases}$ is discontinuous at $x=0$.

Example 4 For what values of a and b is

$$f(x)=\begin{cases} -2, & x\le -1; \\ ax+b, & -1<x<1; \\ 3, & x\ge 1. \end{cases}$$

continuous at every point x?

② Continuous Functions

We now describe the continuity behavior of a function throughout its entire domain, not only at a single point. We define a continuous function to be one that is continuous at every point in its domain. This is a property of the function. A function always has a specified domain, so if we change the domain then we change the function, and this may change its continuity property as well. If a function is discontinuous at one or more points of its domain, we say it is a discontinuous function.

Some continuous functions:

(a) $f(x)=a_0+a_1x+a_2x^2+\cdots+a_nx^n$.

(b) $f(x)=\dfrac{g(x)}{h(x)}$, $g(x)$ and $h(x)\neq 0$ is polynomial functions.

(c) $f(x)=x^a$, $a\in\mathbb{R}$.

(d) $f(x)=a^x$, $a>0$, and $a\neq 1$.

(e) $\sin x$, $\cos x$, $\tan x$, $\cot x$, $\sec x$, $\csc x$.

(f) $f(x)=\sqrt{g(x)}$, $g(x)\ge 0$ is a polynomial function.

Note 3 If $f(x)$ and $g(x)$ are continuous functions, then

$$f(x) \pm g(x),\ f(x)g(x) \text{ and } \frac{f(x)}{g(x)}$$

are continuous functions.

Example 5 Show that function $f(x) = |x|$ is continuous.

Note 4 If $f(x)$ is continuous at $x = c$, then $\lim\limits_{x \to c} f(x) = f(c)$.

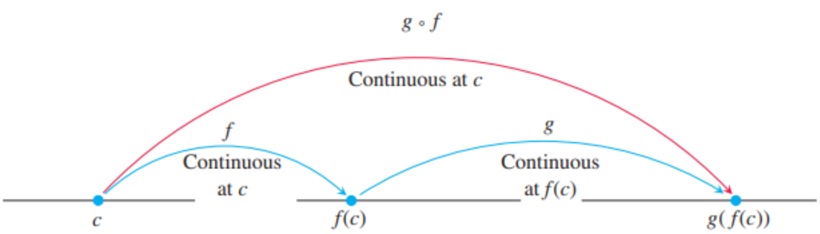
Example 6 Find $\lim\limits_{x \to 1} \sqrt{2x^3 - 1}$.

③ Composites and Continuity

- All composites of continuous functions are continuous.

The idea is that if $f(x)$ is continuous at $x = c$ and $g(x)$ is continuous at $x = f(c)$, then $g \circ f$ is continuous at $x = c$. In this case, the limit as $x \to c$ is $g(f(c))$. That is,

$$\lim_{x \to c} (g \circ f)(x) = \lim_{x \to c} g(f(x)) = g(f(c)).$$



Example 7 Show that the following functions are continuous on their natural domains.

(a) $y=\sqrt{x^2-2x-5}$ (b) $y=\dfrac{x^{\frac{2}{3}}}{1+x^4}$

(c) $y=\left|\dfrac{x-2}{x^2-2}\right|$ (d) $y=\left|\dfrac{x\sin x}{x^2+2}\right|$

Example 8 Find the following limits.

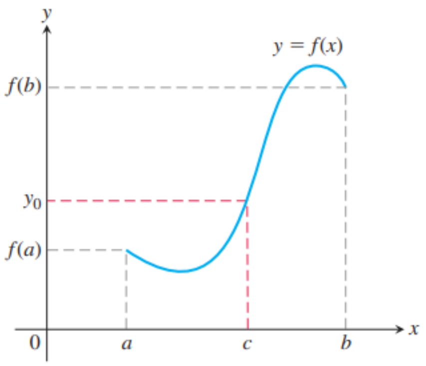
(a) $\lim\limits_{x\to\frac{\pi}{2}}\cos\left(2x+\sin\left(\dfrac{3\pi}{2}+x\right)\right)$;

(b) $\lim\limits_{x\to 0}\sqrt{x+1}\,e^{\tan x}$.

④ Intermediate Value Theorem for Continuous Functions

A function is said to have the Intermediate Value Property if whenever it takes on two values, it also takes on all the values in between.

THEOREM If f is a continuous function on a closed interval $[a, b]$, and if y_0 is any value between f(a) and f(b), then $y_0 = f$(c) for some c in $[a, b]$.



Note 5 We call a solution to $f(x) = 0$ a root or zero of $f(x)$. The Intermediate Value Theorem tell us if $f(x)$ is continuous on $[a, b]$, and $f(a)f(b) < 0$, then $f(x)$ has a zero in (a, b).

Example 9 Show that there is a root of the equation $x^3 - x - 1 = 0$ between 1 and 2.

Example 10 Use the Intermediate Value Theorem to prove that the equation

$$\sqrt{2x+5} = 4 - x^2$$

has a solution.

⑤ Continuous Extension to a Point

If $f(c)$ is not defined, but $\lim_{x\to c} f(x) = L$ exists, we can define a new function $F(x)$ by

$$F(x) = \begin{cases} f(x), & x \neq c; \\ L, & x = c. \end{cases}$$

The function F is continuous at $x = c$. It is called the continuous extension of f to $x = c$.

Example 11 Show that

$$f(x) = \frac{x^2 + x - 6}{x^2 - 4}, \ x \neq 2$$

has a continuous extension to $x = 2$, and find the extension.